INCH-POUND
A-A-58000C
February 18, 2021
SUPERSEDING
A-A-58000B
February 25, 2020

COMMERCIAL ITEM DESCRIPTION

TANK, STORAGE, LIQUID NITROGEN-OXYGEN, CRYOGENIC, HORIZONTAL
1,500, 3,000, AND 6,000 GALLON

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This commercial item description (CID) describes three sizes and two types of skid mounted horizontal storage tank capable of storing and delivering liquid oxygen or liquid nitrogen.

2. CLASSIFICATION. The storage tanks shall be of the following types and sizes. The type and size of the storage tank to be furnished shall be as specified in the acquisition documents.

Type I – Liquid oxygen
Type II – Liquid nitrogen

Size A – 1,500 gallon
Size B – 3,000 gallon
Size C – 6,000 gallon

3. SALIENT CHARACTERISTICS.

3.1 Storage tank description. The tank shall be a complete self-contained, liquid oxygen or liquid nitrogen storage unit consisting of an inner and outer tank with a vacuum maintained in the annular space between the tanks. The system shall include all parts, controls, instruments, and accessories designed for:

a. Filling with liquid oxygen or liquid nitrogen at not less than the flow rates as specified herein.

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b. Storing the liquid at an evaporation rate not greater than as specified herein until needed, and at that time transferring the liquid into smaller containers at a rate not less than as specified herein.

3.1.1 Capacity. The storage tanks shall have a net volume of not less than 1,500 gallons (size A), 3,000 gallons (size B), and 6,000 gallons (size C). The tank shall have a gross water volume rating of not less than 102 percent of the rated net volume.

3.1.2 Capabilities. The tank shall be capable of accepting from commercial transport vessels equipped with transfer pump, storing, and delivering liquid oxygen or liquid nitrogen to military equipment. The tank shall be designed, fabricated, tested, and stamped per latest edition of ASME Boiler and Pressure Vessel Code, Section VIII, Division 1 and Compressed Gas Association (CGA) publications, for a maximum allowable working pressure of not less than 55 pounds per square inch (psi). The tank shall incorporate an integral pressure building unit with heat absorption characteristics resulting in pressure build up rates not less than as specified herein. The tank shall have a skid support structure to enable movement, while empty, by either being dragged over smooth concrete surfaces by a tug, or by being lifted and moved by a forklift. Overall size of the skid base shall be such that the weight of the storage tank assembly is distributed over an area not less than the vertical projection of the outer shell outline.

3.1.3 Annular space insulation. The annular insulation space shall be filled with an insulation that will, together with evacuation of the annular space, ensure compliance with the requirements specified in 3.7.1. The insulating material shall be of a composition commonly used for commercial tanks or better. Application of the insulation shall ensure against settling or displacement in service.

3.1.4 Piping and connections. The tank shall have plumbing lines and associated valves to include, as a minimum, top and bottom fill, liquid withdrawal, vent, full try-cock, pressure building unit, and instrumentation. A control cabinet or housing shall be provided to protect plumbing, valves, and instrumentation from physical and environmental damage.

3.1.4.1 Top and bottom fill line. The top and bottom fill line shall incorporate a 2-inch filter, in accordance with Air Force (AF) Drawing 8518507, upstream of the fill line valve. The filter shall be incorporated into the plumbing system to facilitate removal and replacement. The top and bottom fill line shall terminate in a Standard Connection, OX-150 or NI-150, in accordance with CGA Pamphlet V-6, Standard Cryogenic Liquid Transfer Connections. A fill line safety valve, a fill line drain valve, and a fill line check valve shall also be required.

3.1.4.2 Liquid withdrawal line. The liquid withdrawal line shall terminate in a 1-inch liquid oxygen or liquid nitrogen male ordnance coupling assembly, in accordance with Ordnance Corps Drawings 8991173 for liquid oxygen or 8991171 for liquid nitrogen and shall incorporate a 2-inch filter in accordance with AF Drawing 8518507, located upstream of the liquid withdraw line valve. A withdrawal line safety valve and a withdrawal line drain valve shall also be required.

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3.1.4.3 Liquid fill-drain line. The liquid fill-drain line shall be sized and installed to provide for liquid oxygen flow or transfer into or out of the liquid storage tank assembly inner shell in accordance with 3.7.2.

3.1.4.4 Liquid servicing hose. A durable non-vacuum insulated, flexible stainless steel cryogenic transfer hose in accordance with commercial standards and terminating in a 1-inch liquid oxygen or liquid nitrogen female coupling in accordance with Ordnance Corps Drawing 8991174 for liquid oxygen or 8991172 for liquid nitrogen on each end shall be provided with the tank. Storage provisions for the hose assembly shall also be provided on the tank assembly for transportation purposes.

3.1.4.5 Vapor vent line. The vapor vent line shall permit filling the liquid storage tank assembly through the liquid fill-drain line per the requirements specified in 3.7.4. The vapor vent shall be designed to allow vapor to exit the exterior of the control cabinet and as low as possible to the ground avoiding all contact with outer vessel.

3.1.4.6 Vacuum seal-off valve. The outer shell shall have a vacuum seal-off valve that will serve as the vacuum pump-out connection for evacuation of the annular space, an isolation valve, and the outer shell safety relief device and shall be located within the control cabinet near the bottom.

3.1.5 Pressure buildup system. The pressure buildup system shall operate by vaporizing a portion of the liquid contained in the liquid storage tank assembly inner shell and pressurizing the inner shell vapor space with the resultant vapors. Vaporization of the liquid shall be accomplished in an atmospheric air heated coil that forms a direct external connection between the inner shell liquid and vapor phases through the pressure control system. Pressure buildup rate shall be as specified in 3.7.5.

3.1.6 Controls, instruments, and accessories. All automatic and manual controls, isolation valves, fittings, safety devices, and accessories needed for safe and efficient operation of the tank shall be provided.

3.1.6.1 Instrumentation. All instruments necessary for effective and safe operation of the tank shall be provided. Specific marking requirements for the pressure gage and liquid level gage are as follows.

3.1.6.1.1 Liquid level gage. The liquid level gage shall be scaled to indicate gallons of liquid. The scale shall be of the dual reading type, capable of reading liquid oxygen and liquid nitrogen. (This dual reading gage is required regardless of the type tank specified). The outer band shall indicate and be marked for oxygen. The inner band shall indicate and be marked for nitrogen. Both bands, inner and outer, shall be green from 0 gallons to the net volume of the size tank specified and shall be red from the net volume to the gross volume of the size tank specified. The scale range shall be from 0 to the gross volume of oxygen for the size tank specified. The scale shall have graduations every 200 gallons with 50 gallon increments. The gage face shall also be marked in red lettering, “USE NO OIL.” Instrumentation lines shall be fabricated using seamless stainless steel tubing.

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3.1.6.1.2 Pressure gage. The pressure gage provided shall have a range from 0 to 100 psi. The gage shall have a green color band from 0 to 50 psi and a red color band from 50 to 100 psi. The scale shall have graduations every 10 psi with 1 psi increments. The gage face shall also be marked in red lettering, "USE NO OIL." A tap into the pressure gage line shall be provided for attachment and use of external test equipment. The tap shall terminate in a bulkhead union (Nipple, Tube) or equivalent. The fitting shall be supported in an area adjacent to the pressure gage and shall be sealed with a tethered AN929-4 cap. Instrumentation lines shall be fabricated using seamless stainless steel tubing.

3.1.6.1.3 Vacuum sensing tap. A vacuum sensing tap shall be installed into the annular space, terminating in a 1/8-inch female pipe thread fitting, shall be provided for attachment and detachment of vacuum sensing equipment without degrading the vacuum in the annular space.

3.1.7 Protection devices. Pressure relief devices shall be provided for the protection of the inner tank and the outer shell. The inner tank shall have, as a minimum, a primary and secondary relief device sized and set per ASME Code, section VIII, Division 1 and CGA P-40 and CGA P-59 to allow for a normal operating pressure of 55 psi. Contractor shall provide recommended inspection and test frequencies for the pressure relief devices per 2.5.8 of the NBIC Part 2 Inspection Code.

3.2 Design and construction. The design shall promote cost effective, life-cycle sustainability by addressing considerations such as incorporating open standards, reducing pollutant emissions and wastes, while satisfying system performance requirements. It shall be designed and constructed so that no parts will work loose in service, and to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. It shall be designed so that all pressure relief devices, top and bottom fill line, liquid withdraw line connections, vapor vent line, and full tri-cock ports are positioned to prevent liquid nitrogen/oxygen contacting the outer vessel. It shall be weatherproof and designed to prevent the intrusion of water, sand, and dust into critical operating components. The storage tanks shall be new and of the manufacturer's commercial design in effect on the date of the solicitation. The tanks shall meet the current requirements specified herein and shall conform to current commercial industrial practice.

3.2.1 Materials, protective coatings, finish, and water exclusion.

3.2.1.1 Materials. Materials that deteriorate when exposed to sunlight, weather, or operational conditions normally encountered during the service life of the item shall not be used or shall have means of protection against such deterioration that does not prevent compliance with the performance requirements specified herein.

3.2.1.2 Protective coatings. Protective coatings that chip, crack, or scale with age or extremes of climatic conditions or when exposed to heat shall not be used. Exposed surfaces of fasteners, handles, and fittings shall also be primed and painted. See 3.2.4.1 for prohibited hazardous materials.

3.2.1.3 Finish. The exterior finish color of the storage tank and the inner surfaces of compartments shall be Gloss White, color number 17925 of SAE AMS-STD-595.

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3.2.1.4 Exclusion of water. The design of the storage tank shall be such as to prevent water leaking into, or being driven into, any part of the storage tank interior. All doors, panels, covers, etc., shall be provided with sealing arrangements such that the entry of water is minimized when these items are correctly closed. Sharp corners and recesses shall be avoided so that moisture and solid matter cannot accumulate to initiate localized attack. Sealed floors with suitable drainage shall be provided for storage compartments, and other areas in the storage tank that could collect and retain water.

3.2.1.4.1 Fluid traps and faying surfaces. There shall be no fluid traps on the storage tank. Faying surfaces of all structural joints, except welded joints, shall be sealed to preclude fluid intrusion.

3.2.1.4.2 Ventilation. Ventilation shall be sufficient to prevent moisture retention and buildup.

3.2.1.4.3 Drainage. Drain holes shall be provided to prevent collection or entrapment of water or other unwanted fluid in areas where exclusion is impractical. All designs shall include considerations for the prevention of water or fluid entrapment and ensure that drain holes are located to effect maximum drainage of accumulated fluids. The minimum size of the drain holes shall be 0.25 inch.

3.2.2 Markings. All external devices which require an operational or maintenance interface shall be marked in accordance with MIL-STD-130. Markings shall be applied with decals and shall be 1-inch high block letters unless prohibited by the available space. In such cases, the markings shall be the largest size possible. Markings, Information/Caution shall be Lusterless Black, color number 37038 of SAE AMS-STD-595, and Markings, Warning/Danger shall be Lusterless Red, color number 31136 of SAE AMS-STD-595. The tank shall be labelled per NFPA 55 and NFPA 704. The center of gravity of the storage tank shall be stenciled on the unit within 1-inch of the calculated center of gravity. The storage tank nameplate 8752448-01 (see 3.2.3.1) shall be securely attached to the tank in a readily visible location. Each tank shall require decals, 8221044-01 for liquid oxygen and 8221045-01 for liquid nitrogen applied one on each side of the tank at the middle above the horizontal center line of the tank, and one centered on each end of the tank.

3.2.3 Identification and information plates.

3.2.3.1 Identification plate. An identification plate in accordance with MIL-STD-130 shall be securely attached to the storage tank in a readily accessible location. The identification plate shall contain the following information: nomenclature, part number, serial number, date of manufacture, manufacturer's name, Commercial and Government Entity (CAGE) code, date of warranty expiration, and National Stock Number (NSN). The storage tank and any of its components for which the Government's unit cost is more than $5,000, is serially managed, or the procuring agency determines is mission essential, shall have Unique Identification (UID) (also known as Item Unique Identification (IUID)) information permanently affixed on or near the respective identification plate(s), marked in accordance with MIL-STD-130. UID information shall be included as both a bar code and human readable markings.

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3.2.3.2 Transportation data plate. A transportation data plate shall be securely attached to the storage tank in a readily accessible location. The plate shall contain at least the following information:

a. Side and rear silhouette views of the storage tank.

b. Horizontal and vertical location of the center of gravity of the storage tank in transportable configuration, marked on the silhouette views.

c. Shipping weight.

d. Loading cubage.

e. Overall height, width, and length.

f. Tie down information.

3.2.3.3 Lifting instruction plate. An instruction plate shall be furnished containing all information necessary for transportation personnel including but not limited to safely lift and move the empty tank with forklift trucks of specified capacity and safely lift the empty tank with a 2-g acceleration. The plate shall be attached to the outside of the control housing assembly near the transportation data plate. The lifting instruction plate shall be of the same material and prepared in the same manner as the transportation data plate.

3.2.4 Environment, Safety, and Occupational Health (ESOH).

3.2.4.1 Hazardous material. The design shall minimize and control hazards associated with the inclusion or use of hazardous or toxic materials and the generation of toxic or noxious gases. The storage tank shall not generate or use Class I or Class II Ozone Depleting Substances (ODS) during operation, maintenance, or disposal. The storage tank shall not contain or use hexavalent chromium. The storage tank shall not contain or use cadmium without written approval by the procuring activity. For the purposes of this requirement, the Class I ODS and controlled substances identified in chapter 4 of AFI 32-7086 shall not be used in any system, component, or process.

3.2.4.2 Component protection. All space in which work is performed during operation, service, and maintenance shall be free of hazardous protrusions, sharp edges, or other features which may cause injury to personnel. All tubes and hoses shall be supported and protected to minimize chafing and abrasion. Grommets shall be provided wherever tubes or hoses pass through bulkheads, partitions, or structural members.

3.2.4.3 Foreign object damage (FOD). All loose metal parts, such as pins or connector covers, shall be securely attached to the storage tank with wire ropes or chains. "Dog tag" style beaded chains shall not be provided. Removable panels, if provided, shall be attached with captive fasteners.

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3.2.6 Human systems integration. To the maximum extent possible the storage tank shall be designed in accordance with MIL-STD-1472 for ease of operation, inspection, and maintenance, including the use of arctic mittens.

3.2.7 Fastening devices. All screws, bolts, nuts, pins, and other fastening devices shall be properly designed, manufactured, and installed with adequate means of preventing loss of torque or adjustment. Cotter pins, lock-washers, or nylon patches shall not be used for this purpose, except for the attachment of trim items or as provided in commercial components. Tapped threads shall have a minimum thread engagement in accordance with table I.

TABLE I. Minimum thread engagement.

Material	**Minimum Thread Engagement**
Steel	1.0 times the nominal fastener diameter
Cast iron, brass, or bronze	1.5 times the nominal fastener diameter
Aluminum, zinc, or plastic	2.0 times the nominal fastener diameter

3.2.8 Welders and welding. The contractor shall make available to the Government certifications for all welders being utilized on the storage tank. The surface parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the welded parts. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof, and service loadings.

3.2.8.1 General welding. All welders shall be certified to weld in accordance with AWS D1.1 and AWS D1.2, as applicable. With the exception of 3.2.8.2, welding procedures and all welding on the storage tank shall be in accordance with AWS D1.1 and AWS D1.2, as applicable.

3.2.8.2 Inner and outer tank welding. All inner and outer tank welding shall be accomplished by welders certified in accordance with ASME Section IX of the Boiler and Pressure Vessel Code as stated in ASME Section VIII, Div. 1, UG-9.

3.3 Environmental conditions. The tank shall operate in an unsheltered environment during weather conditions encountered in commercial applications.

3.4 Dimensions. The overall dimensions of the tank including plumbing (see 3.5.1) shall not exceed the values given below:

Tank Size (gallons)	Length (inches)	Width (inches)	Height (inches)
1,500	225	80	95
3,000	260	108	120
6,000	450	108	120

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3.5 Transportability.

3.5.1 Surface transportability. The storage tank shall be transportable via all modes of surface shipment (highway, rail, and water), and shall be capable of withstanding the mechanical shock and vibration characteristics of highway, rail, and water transport, except that design for rail impact is not required.

3.5.2 Surface mobility. The storage tank shall have a skid support structure to enable movement, while empty, by either being dragged over smooth concrete surfaces by a tug, or by being lifted and moved by a forklift. Overall size of the skid base shall be such that the weight of the storage tank assembly is distributed over an area not less than the vertical projection of the outer shell outline.

3.6 Maintainability. Parts and assemblies shall be located and mounted to provide clearance for repair and other maintenance and, where practicable, to permit removal and replacement of any part or assembly by removing or disconnecting only mounting bolts or fasteners, panel doors, tubing, or control cable to the part or assembly being removed.

3.6.1 Preventive maintenance. The recommended preventive maintenance interval (PMI) shall be at least 50 servicing cycles during which 10 gallons of liquid are transferred to a suitable container connected to the transfer hose. A servicing cycle shall consist of pressurizing the liquid storage tank assembly inner shell vapor space, transferring liquid into another vessel, and depressurizing the inner shell vapor space back to atmospheric pressure.

3.6.2 Inspection and servicing provisions.

a. Pre-use inspections and servicing tasks shall not require tools.

b. Drain plugs and filters shall be directly accessible from the ground and oriented to have unimpeded drainage to a catch pan.

3.6.3 Special tools. The design of the item shall minimize the requirement for special tools (see 7.3.2). All special tools shall be provided with, and stored on the storage tank.

3.6.3.1 Vacuum gauge and vacuum gauge tube. A Teledyne Hastings HPM 4/5/6 Vacuum gauge and Teledyne Hastings DV-6 series vacuum gauge tube shall be provided as a special tools.

3.7 Performance.

3.7.1 Insulation system performance. The insulation system shall meet the following performance requirements.

3.7.1.1 Heat leak evaporation rate. The tank heat leak evaporation loss rate shall not exceed the Nominal Evaporation Rate (NER) given below under the following conditions:

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a. The tank inner shell is filled to not less than 50 percent of design capacity with liquid oxygen or liquid nitrogen at atmospheric pressure.

b. The tank is in a location where the ambient temperature is not less than 125 °F.

c. The upper surface of the tank is continuously exposed to solar or equivalent heat radiation of 100 to 120 watts per square foot for not less than 8 hours each day.

d. Not less than 30 days have elapsed since evacuation of the tank annular space.

e. The tank has been subjected to all tests specified herein.

Size, gallons	NER, %/day Oxygen	NER, %/day Nitrogen
1,500	0.62	0.99
3,000	0.35	0.57
6,000	0.24	0.38

3.7.1.2 Vacuum retention. The tank annular space absolute pressure shall not rise above 35 microns Mercury (Hg) or show any increase attributable to leakage or outgassing within 7 days (168 hours) when:

a. Prior to the 7-day test period, the tank annular space is evacuated to ensure an absolute pressure of equal or less than 35 microns Hg within 24 hours after the warm tank is filled with liquid oxygen or liquid nitrogen to 50 percent or more of design capacity.

b. The vacuum line is sealed after evacuation to prevent further evacuation prior to completion of the 7-day test period and the final absolute pressure measurement.

c. The vacuum retention testing specified under 5.5.1.2 is accomplished during the 7-day test period.

d. The tank inner shell is pressurized to not less than 30 psi with oxygen or nitrogen gas throughout the 7-day test period.

e. The tank is subjected to ambient temperature throughout the test period.

3.7.2 Liquid fill and drain line performance. Pressure drop across the liquid fill/drain line shall not exceed 5 psi at a rate of 100 gallon per minute (gpm).

3.7.3 Liquid service line performance. The service line shall deliver liquid at 25 gpm without the pressure drop across the line and filter exceeding 5 psi.

3.7.4 Vapor vent line performance. The vapor vent line shall permit filling the liquid storage tank assembly through the liquid fill-drain line at any rate up to and including 100 gpm without the inner shell vapor phase pressure exceeding 15 psi when the liquid storage tank assembly is cold.

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3.7.5 Pressure buildup rate. The pressure buildup system shall pressurize the inner shell vapor space to 50 psi when the tank contains 150 gallons or more of liquid oxygen/liquid nitrogen. The buildup system shall pressurize the tank inner shell vapor space to 50 psi within 20 minutes after closure of the vapor vent line shutoff valve and initiating buildup when the tank is filled to between 15 to 20 percent of rated capacity, and has been vented to atmospheric pressure for not less than 4 hours.

3.8 Workmanship. All parts of the tank shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to the following:

a. Freedom from blemishes, defects, burrs, and sharp edges.

b. Accuracy of dimensions, radii of fillets, and marking of parts and assemblies.
c. Thoroughness of soldering, welding, brazing, painting, and riveting.

d. Thorough removal of rust, slag, flux, and other foreign materials from the inside of the liquid storage tank assembly and from all piping, valve, connection, filter housing, and other surfaces that contact oxygen/nitrogen product.

e. Alignment of parts and tightness of assembly screws, bolts, rivets, etc.

3.9 General cleaning instructions. Following completion of fabrication and assembly operations, the storage tank shall be thoroughly cleaned and degreased to remove dirt; excess soldering, brazing, and welding flux; welding slag, scale, metal chips, loose or chipped paint, spilled chemicals, and other foreign materials. Cleaning and degreasing shall be performed on surfaces contacting oxygen in accordance with SAE ARP-1176 and to meet the following requirements:

a. When the storage tank is filled to 90 percent or more of design capacity with filtered liquid oxygen or nitrogen and allowed to set for not less than 2 hours, the residue remaining after evaporation of the first liter of liquid withdrawn through the fill drain line, with its filter removed, shall contain (see 5.5.1.5):

(1) No solid particles with any dimension greater than 1,000 microns.
(2) No fibrous particle with a length greater than 6,000 microns.
(3) No more than 25 milligrams total of both solid and fibrous particles.

b. When the liquid remaining in the tank after withdrawal of the above 1 liter sample is discharged through the liquid fill-drain line and installed filter at a rate not less than 150 gallons per minute, the total amount of contaminant collected on the filter shall not exceed 0.5 gram (see 5.5.1.6).

c. Storage tank surfaces, fittings, and other parts that will be in contact with oxygen shall be degreased in accordance with SAE ARP-1176. Precautions shall be taken to ensure that solvents do not contact parts fabricated from incompatible materials. Petroleum and other flammable solvents shall not be used on such surfaces. All cleaning materials

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and solvents shall be thoroughly removed from the tank components following the cleaning and degreasing process.

4. REGULATORY REQUIREMENTS.

4.1 Recycled recovered materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life-cycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt, or refurbished items shall not be provided.

4.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of Environmental Protection Administration (EPA) designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

5. PRODUCT CONFORMANCE PROVISIONS.

The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace, modified as necessary to comply with the requirements herein. The Government reserves the right to require proof of such conformance.

5.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First production inspection (see 5.2).

b. Conformance inspection (see 5.3).

5.2 First production inspection. The first production storage tanks shall be subjected to the analyses, demonstrations, examinations, and tests described in 5.5.1 through 5.5.9. The contractor shall provide or arrange for all test equipment and facilities.

5.3 Conformance inspection. Each production storage tank shall be subjected to the examination described in 5.5.1.

5.4 Inspection requirements.

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5.4.1 General inspection requirements. Apparatus used in conjunction with the inspections specified herein shall be laboratory precision type, calibrated at proper intervals to ensure laboratory accuracy.

5.4.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data or shorter intervals shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time finished.

d. Ambient temperature.

5.4.3 Test rejection criteria. Throughout all tests specified herein, the storage tank shall be closely observed for the following conditions, which shall be cause for rejection.

a. Failure to conform to design or performance requirements specified herein.

b. Any spillage or leakage of any liquid.

c. Structural failure of any component, including permanent deformation, or evidence of impending failure.

d. Evidence of excessive wear. If excessive wear is suspected, the original equipment manufacturer's (OEM's) specifications or tolerances shall be utilized for making a determination.

e. Evidence of corrosion or deterioration.

f. Misalignment of components.

g. Conditions that present a safety hazard to personnel during operation, servicing, or maintenance.

5.5 Test methods.

5.5.1 Examination of product. Each storage tank shall be examined to verify compliance with the requirements herein prior to accomplishing any other demonstrations or tests listed in 5.5. A contractor-generated, Government-approved checklist (part of the test procedure) shall be used to identify each requirement not verified by an analysis, certification, demonstration, or test, and shall be used to document the examination results. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and sealants, and their application, welding, fastening, and markings. Proper operation of each storage tank function shall be verified.

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Certifications and analyses shall be provided in accordance with table II. Each production storage tank shall be inspected to a Government-approved reduced version of the checklist.

5.5.1.1 Pressure test. Each component and circuit of the tank that operates under positive pressure in normal service shall be pressure tested in accordance with section VIII of the ASME Boiler and Pressure Vessel Code. Any distortion, damage, or leaks resulting from the pressure testing shall be cause for rejection of the components affected.

5.5.1.2 Vacuum retention test.

a. The liquid storage tank assembly inner shell shall be filled with liquid oxygen or liquid nitrogen to not less than 60 percent of design capacity and permitted to temperature stabilize. Following temperature stabilization of the liquid storage tank assembly and with the inner shell vapor space at approximately atmospheric pressure, the annular insulation space absolute pressure shall be determined. The annular insulation space absolute pressure shall not exceed 35 microns Hg. The inner shell vapor space shall then be pressurized to not less than 30 psi with oxygen or nitrogen gas obtained by vaporizing part of the contained liquid with the pressure buildup system, and the tank allowed to set for not less than 7 days (168 hours) filled and pressurized.

b. Following completion of the test period the inner shell vapor space pressure shall be reduced to atmospheric pressure and the annular insulation space absolute pressure measured. The annular insulation space absolute pressure shall not exceed 35 microns Hg or show any increase attributable to leakage or outgassing, or the tank shall be considered to have failed the test.

TABLE II. Certifications and analyses.

Paragraph	**Required Certifications and Analyses**
3.1.2 Capabilities.	Inner and outer tank certification to ASME Boiler and Pressure Vessel Code, Section VIII, Division 1.
3.1.7 Protection devices.	Contractor certification pressure relief devices tested to 55 psi per ASME Boiler and Pressure Vessel Code, Section VIII, Division 1 and CGA P-40 and CGA P-59.
3.2 Design and construction.	Contractor certification product meets all requirements of this CID.
3.2.4.1 Hazardous material.	Contractor certification of compliance.
3.2.8.1 General welding.	Welder's certifications.
3.2.8.2 Inner and outer tank welding.	Welder's certifications.
4.1 Recycled recovered materials.	Contractor statement of compliance.
4.2 Green Procurement Program.	Contractor statement of compliance.

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5.5.1.3 Cleanliness test visual inspection. All equipment, pipes, and components that will contact liquid oxygen/nitrogen shall be visually examined for evidence of corrosion products, metal chips, scale, weld scale, oil, grease, paints, preservatives, decals, or other contamination of foreign matter. Any evidence of contamination or foreign matter shall require cleaning and retest.

5.5.1.4 Ultra violet light inspection. All accessible surfaces and parts that will contact liquid oxygen/nitrogen shall be visually inspected for hydrocarbons with ultraviolet light. Any evidence of fluorescence shall require re-cleaning and retest.

5.5.1.5 Particulate matter. Following the cleaning and degreasing operations specified herein and equipment final assembly, the liquid storage tank inner shell shall be filled to not less than 90 percent of design capacity with liquid oxygen or liquid nitrogen. The storage tank shall then be permitted to set for at least 2 hours undisturbed. Following the 2-hour period, a 2-liter or larger Dewar flask shall be placed under the discharged end of the fill-drain with its filter removed, the fill-drain line shutoff valve opened, and not less than 1-liter of the first liquid discharge through the line collected in the Dewar flask. This sample shall be evaporated to dryness and the residue examined for particulate matter. The liquid storage tank and the associated piping and fittings shall be re-cleaned and this test repeated until the particulate matter remaining as residue contains:

a. No solid particle with any dimension greater than 1,000 microns.

b. No fibrous particles with a length greater than 6,000 microns.

c. Not more than 25 milligrams total of both solids and fibrous particles.

5.5.1.6 Total solids. Following withdrawal of the 1-liter sample specified in 5.5.1.5, the entire load of contained liquid shall be discharged at a flow rate of not less than 150 gpm through the fill-drain line and the filter. The filter shall be thoroughly cleaned and the removed material collected, dried, and weighed after each use. If more than 0.5 gram of contaminant material is collected on the filter during discharge of the liquid from the tank, the liquid storage tank assembly and the associated piping and fittings shall be re-cleaned. Cleaning and testing shall be repeated until the total amount of contaminant collected on the filter from discharging the liquid storage tank assembly load is less than 0.5 gram.

5.5.2 Weight and dimension tests.

5.5.2.1 Weight and center of gravity test. The weight and center of gravity of a first production storage tank shall be measured to demonstrate compliance with the weight and center of gravity requirement of 3.2.2 and 3.2.3.2.

5.5.2.2 Dimension measurement. A first production storage tank shall be measured to demonstrate compliance with the dimensional requirements of 3.4.

5.5.3 Surface transportability analysis and mobility test.

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5.5.3.1 Surface transportability analysis. An engineering analysis shall be performed to demonstrate compliance with 3.5.1. The engineering analysis shall utilize the data for road transportation in accordance with MIL-STD-810, method 514.8, table 514.8C-III.

5.5.3.2 Surface mobility test. Surface mobility of the skid mounted storage tank shall be demonstrated by dragging the empty storage tank not less than 10 feet across a concrete surface. The ability of the empty storage tank to be lifted from a concrete surface and moved by a forklift shall be demonstrated by lifting the empty storage tank, moving it not less than 10 feet and returning it to the concrete surface.

5.5.4 Maintainability demonstration. All recommended preventive maintenance tasks shall be performed and the task times shall be recorded. The recommended frequencies of the preventive maintenance tasks and the times recorded to accomplish the tasks shall be used to develop an expected value of preventive maintenance time per measure of use, such as calendar time or cycles of operation. All preventive maintenance tasks recommended to be performed daily and at the routine PMI shall also be performed by personnel wearing arctic mittens.

5.5.5 Preventive maintenance test. 50 servicing cycles in accordance with 3.6.1 shall be performed. If any maintenance tasks are performed in addition to those listed in the servicing cycle procedure for the test it is a failure.

5.5.6 Capacity. The storage tank volume shall be measured by filling with water. The tank shall have a gross water volume rating of not less than 102 percent of the rated net volume.

5.5.7 Annular space insulation test.

5.5.7.1 Heat leak evaporation rate. The heat leak evaporation loss rate test shall consist of the following individual test operations conducted in the order listed. Throughout the complete test, the inner shell vapor phase pressure shall remain at as near atmospheric as necessary pressure drops through the measuring apparatus wi11 permit.

5.5.7.2 Test environment. Following the vacuum retention test and with the tank remaining not less than 50 percent of design capacity with liquid oxygen or liquid nitrogen, the tank shall be placed in a location where the ambient temperature will remain at not less than 125 °F throughout the periods specified.

5.5.7.3 Inner shell pressure. The inner shell vapor space shall be vented to atmospheric pressure for not less than 24 hours.

5.5.7.4 Liquid loss rate. The tank shall remain under the specified ambient temperature and inner shell vapor phase pressure conditions for an additional 72 hours while heat leak evaporation rate losses for the tank are determined by passing all vapors vented from the inner shell through a recording or totaling-type gas flow-meter. During this 72-hour period, the upper surface of the tank shall be exposed to the full impact of solar or equivalent heat radiation of 100 to 120 watts per square foot for three separate periods of not less than 8 hours each, with each exposure followed by a period of not less than 16 hours during which no radiation is applied. The total weight of

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vapors vented during the test, the weight of vapors vented since the preceding reading, the average ambient temperature to which the tank has been subjected during the preceding period, and the average barometric pressure to which the tank has been subjected during the preceding period shall be recorded at intervals of not to exceed 8 hours during the 72-hour test period, and the average ambient temperature to which the tank was exposed during the 72-hour test period shall be calculated. The evaporation loss rates shall be reported as the pounds of liquid oxygen or liquid nitrogen vaporized per 24-hour day, and shall not exceed the limit specified in 3.7.1, or the tank shall be considered to have failed the test.

5.5.8 Liquid transfer tests.

5.5.8.1 Fill-drain line transfer. Liquid oxygen/nitrogen shall be transferred into the tank through a fill-drain at not less than 100 gpm for a length of time adequate to determine the pressure drop resulting from friction losses in the filter and line. The pressure drop shall not exceed 5 psi, or the tank shall be considered to have failed the test.

5.5.8.2 Servicing line transfer. Liquid oxygen shall be transferred from the tank through the servicing line at not less than 25 gpm for a length of time adequate to determine the pressure drop resulting from friction losses in the filter and line. The pressure drop shall not exceed 5 psi, or the tank shall be considered to have failed the test. Liquid nitrogen may be used at a flow rate which will provide a pressure drop equivalent to that resulting from the specified liquid oxygen flow. If liquid nitrogen is employed for this purpose, data shall be provided to substantiate the flow rate used.

5.5.8.3 Vapor vent line. The storage tank shall be subjected to conditions that will ensure all liquid storage tank assembly components attaining a temperature of not less than 125 °F. After the liquid storage tank components have stabilized at this temperature, liquid oxygen or liquid nitrogen shall be transferred into the inner shell through the fill-drain line until the inner shell has cooled sufficiently to permit filling of 100 gpm rate. The rate of transfer during cool down shall be the maximum that will maintain a liquid storage tank assembly inner shell pressure of less than 15 psi. The cool down time shall not exceed 15 minutes, or the tank shall be considered to have failed the test. Following cool down, the liquid storage tank assembly shall be filled at a rate of not less than 100 gpm. The inner shell vapor phase pressure shall not exceed 15 psi when filled at the specified rate, or the tank shall be considered to have failed the test. Liquid nitrogen may be used at a flow rate which will provide a pressure drop equivalent to that resulting from the specified liquid oxygen flow. If liquid nitrogen is employed for this purpose, data shall be provided to substantiate the flow rate used.

5.5.9 Pressure build rate. The liquid storage tank assembly shall be filled with liquid oxygen or liquid nitrogen between 15 and 20 percent of capacity and permitted to set with the inner shell vapor space vented to atmospheric pressure for not less than 4 hours and until all external components of the tank reach approximately ambient temperature. The inner shell vapor vent line shutoff valve shall then be closed and the pressure buildup system actuated. The length of time required after closing of the vent valve until the vapor phase pressure has reached 50 psi shall be recorded on the test data sheets and shall not exceed 20 minutes, or the tank shall be considered to have failed the test.

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6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Source of documents.

7.1.1 American Society of Mechanical Engineers (ASME) documents. Copies of ASME documents may be obtained from ASME, Two Park Avenue, New York, NY 10016, or online at https://www.asme.org.

7.1.2 American Welding Society (AWS) documents. Copies of AWS documents may be obtained from the American Welding Society, 8669 Northwest 36 Street, Suite 130, Miami, FL 33166-6672, or online at https://www.aws.org.

7.1.3 ASTM documents. Copies of ASTM documents may be obtained from the ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, PA 19428-2959, or online at https://www.astm.org/.

7.1.4 Compressed Gas Association (CGA) documents. Copies of CGA documents may be obtained from the Compressed Gas Association, 14501 George Carter Way, Suite 103, Chantilly, VA 20151, or online at https://www.cganet.com/.

7.1.5 Environmental Protection Agency (EPA). Copies of EPA documents may be obtained from the Environmental Protection Agency, 1200 Pennsylvania Avenue, Washington, DC 20460, or online at https://www.epa.gov.

7.1.6 Federal Acquisition Regulations (FAR). Copies of FARs may be obtained from the Superintendent of Documents, Government Printing Office, Washington, DC 20402, or online at https://www.acquisition.gov/?q=browsefar.

7.1.7 Government documents. Copies of specifications, standards, and handbooks are available online at https://quicksearch.dla.mil/.

7.1.8 Other Government drawings. The following other Government drawings form a part of this document. The drawings describe the Government interface couplings and data plate.

Ordinance Corps

8991171	Assembly, LN2 Coupling, Male Half, 1-Inch
8991172	Assembly, Liquid Nitrogen Coupling, Female Half l-Inch
8991173	Assembly, LOX Coupling, Male Half, 1-Inch
8991174	Assembly, LOX Coupling, Female Half, l-Inch
9051080	Filter Assembly, 1”
9051081	Filter Assembly, 2”

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Air Force

8221044-01	Decal, Liquid Oxygen
8221045-01	Decal, Liquid Nitrogen
8518507	Filter, Assembly, 2-Inch
8752448-01	Nameplate

(Copies of the above drawings required by suppliers in connection with specific acquisition functions shall be obtained from the contracting activity, or as directed by the contracting officer.)

7.1.9 National Board Inspection Code (NBIC). Copies of the NBIC are available from The National Board of Boiler & Pressure Vessel Inspectors. Headquarters, 1055 Grupper Ave., Columbus Oh 43229-1183 or online at http://www.nationalboard.org

7.1.10 SAE International standards. Copies of SAE standards may be obtained from the SAE International, 400 Commonwealth Drive, Warrendale, PA 15096, or online at http://www.sae.org/.

7.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this CID.

b. Type (see 2).

c. Size (see 2).

d. When first production inspection is required.

7.3 Definitions.

7.3.1 Common hand tools. A non-powered tool that is likely to be found in a typical mechanic's toolbox. Common hand tools include open-end, boxed-end, combination, socket (both 6- and 12-point in both standard and deep-well), and hex-key wrenches, in SAE sizes up to and including 1-inch and metric sizes up to and including 25-mm; ratchet handles, extensions, and swivels; slotted- and phillips-head screwdrivers; regular and snap-ring pliers; and a ball-peen hammer.

7.3.2 Special tool. A tool that is not commercially and readily available from a source other than the storage tank contractor.

7.4 Keywords.

LN2
LOX

7.5 Changes from previous issue. Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any

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inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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